S181M116 Derivatives and Alternative Investments

Lecture 1: Introduction, Futures Markets, and Risk Lessons

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Today

1. What derivative contracts are and why they matter.
2. Exchange-traded versus OTC market structure.
3. Forward, futures, and option payoffs.
4. Futures contract design, convergence, margin, and clearing.
5. Hedging, speculation, arbitrage, and why these labels matter.
6. Risk-management lessons from derivatives mishaps.

Core Message

Derivative markets combine three layers

1. **Contract economics:** what future cash flows are promised?
2. **Market plumbing:** who guarantees performance, posts collateral, and settles gains/losses?
3. **Pricing logic:** can the payoff be replicated or bounded by no-arbitrage?

A student who knows only the formula misses the institution. A student who knows only the institution misses the valuation logic.

What Is a Derivative?

Definition

A derivative is a contract whose value depends on an underlying variable.

- The underlying can be a stock, bond, exchange rate, commodity, index, interest rate, credit event, or weather variable.
- The contract specifies cash flows at one or more future dates.
- A simple representation is

$$\text{payoff at } T = g(X_T),$$

  where X_T is the underlying variable at maturity.

Why Derivatives Matter

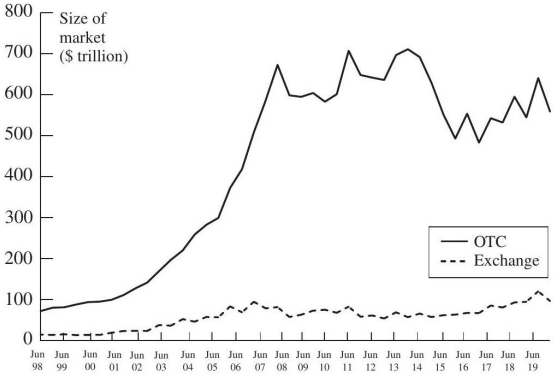
Useful

- Transfer risk.
- Lock in future prices.
- Complete markets.
- Improve price discovery.
- Create tailored exposures.

Dangerous when misused

- Embedded leverage.
- Margin and liquidity pressure.
- Model error.
- Counterparty exposure.
- Weak governance.

Market Size



- Notional measures scale, not current value-at-risk.
- Gross market value is smaller, but notional matters for leverage and interconnectedness.

Payoff, Profit, and Value

Payoff	Cash flow at maturity, usually as a function of S_T.
Profit	Payoff minus initial cost and financing costs.
Value	Market value of the contract today.

Example

A call option payoff may be positive at maturity, but the buyer's profit also subtracts the premium paid upfront.

We will be precise about these distinctions throughout the course.

Exchange-Traded Derivatives

- Contracts are standardized by the exchange.
- Trading is anonymous and usually electronic.
- The clearing house stands between buyers and sellers.
- Credit risk is controlled through margin, daily settlement, and default funds.

Key idea

Standardization reduces customization but improves liquidity and clearing.

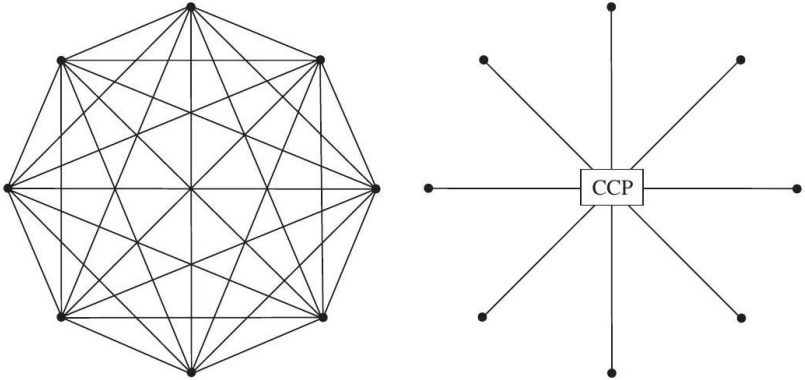
OTC Derivatives

- OTC contracts are privately negotiated.
- Maturity, notional, payoff, collateral, and settlement terms can be customized.
- Many standardized OTC trades are now centrally cleared.
- Nonstandard trades can remain bilateral, governed by collateral and close-out agreements.

Trade-off

OTC flexibility comes with valuation, liquidity, legal, and counterparty-risk complications.

Bilateral Clearing Versus CCP Clearing



(a) (b)

Counterparty Risk and Systemic Risk

Counterparty risk

The risk that the other side of a contract fails to perform.

Systemic risk

The risk that one institution's failure causes losses or funding stress elsewhere in the system.

- OTC networks can transmit distress.
- CCPs reduce bilateral exposure but concentrate risk-management responsibility.
- Collateral reduces credit risk but can create liquidity pressure.

Forward Contract

Definition

A forward contract is an agreement to buy or sell an asset at future date T for delivery price K.

Long forward

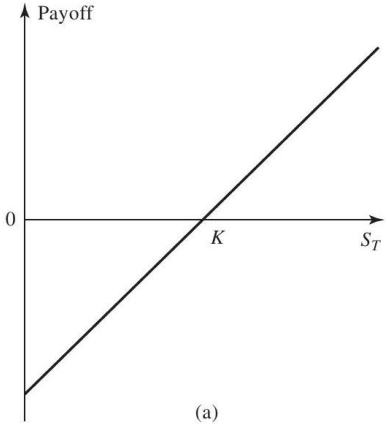
- Agrees to buy at K.
- Benefits when $S_T > K$.

Short forward

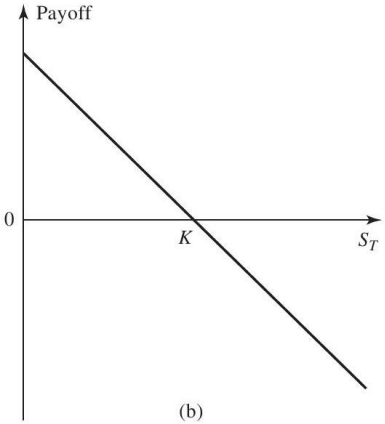
- Agrees to sell at K.
- Benefits when $S_T < K$.

Forward Payoffs

Long payoff: $S_T - K$



Short payoff: $K - S_T$



Forward Hedge Example

A U.S. firm must pay GBP 1 million in six months.

- Six-month forward quote: USD 1.2230 per GBP.
- The firm enters a long GBP forward.
- It locks in a dollar payment of

$$1{,}000{,}000 \times 1.2230 = \$1{,}223{,}000.$$

If $S_T = 1.3000$

Forward payoff is

$$(1.3000 - 1.2230) \times 1{,}000{,}000 = \$77{,}000.$$

Forward Price: First No-Arbitrage Preview

For a non-dividend-paying asset:

$$F_0 = S_0 e^{rT}.$$

Why?

Buying the asset today and financing it must cost the same as agreeing today to buy it later.

- If $F_0 > S_0 e^{rT}$: borrow, buy the asset, short the forward.
- If $F_0 < S_0 e^{rT}$: short the asset, invest proceeds, long the forward.

Cash-and-Carry Arbitrage

Suppose $S_0 = 60$, $r = 5\%$ simple annual rate, and $T = 1$. Then the no-arbitrage forward price is

$$F_0 = 60(1.05) = 63.$$

If the quoted forward price is 67

1. Borrow 60.
2. Buy the asset.
3. Short the forward at 67.
4. At maturity, deliver asset, receive 67, repay 63.

Locked-in profit: $67 - 63 = 4$.

Futures Contract

A futures contract is similar to a forward contract economically, but not institutionally.

Feature	Forward	Futures
Venue	OTC	Exchange
Terms	Customized	Standardized
Counterparty	Bilateral/CCP	Clearing house
Settlement	Usually maturity	Daily
Initial cash flow	Usually none	Margin

Contract Specification

A futures exchange must define:

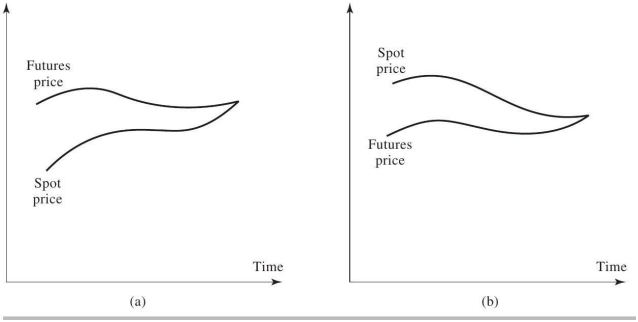
- underlying asset and acceptable grades;
- contract size;
- delivery months;
- delivery location and delivery process;
- price quotation convention;
- tick size;
- price limits and position limits.

Why so much detail?

Standardization makes the contract tradable by many participants who do not know each other.

Convergence

As delivery approaches, the futures price converges to the spot price.



Anchor

Even when delivery is rare, the possibility of delivery ties the futures market to the cash market.

Margin Is Collateral

Margin is not the price of the futures contract

It is collateral posted to protect the clearing system against default.

- **Initial margin:** amount deposited when the position is opened.
- **Maintenance margin:** minimum acceptable balance.
- **Margin call:** request to restore the account after losses.
- **Variation margin:** daily settlement of gains and losses.

Daily Settlement

Let N be the number of contracts and Q the contract size.

Long futures daily P&L

$$\mathrm{P\&L}_t = NQ(F_t - F_{t-1}).$$

Short futures daily P&L

$$\mathrm{P\&L}_t^{short} = NQ(F_{t-1} - F_t).$$

Futures gains and losses are realized daily, not merely at maturity.

Margin Example

A trader is long two gold futures contracts. Each contract is for 100 ounces.

- Total exposure: $2 \times 100 = 200$ ounces.
- Yesterday's settlement: $F_{t-1} = 1{,}750$.
- Today's settlement: $F_t = 1{,}741$.

$$\mathrm{P\&L}_t = 200(1{,}741 - 1{,}750) = -1{,}800.$$

Interpretation

The long loses USD 1,800 today. The short gains USD 1,800 today.

Why Futures Can Create Liquidity Risk

- Daily settlement reduces credit risk to the clearing house.
- But it forces losing traders to fund losses immediately.
- A trader can be correct about the final price and still fail because interim margin calls are too large.

Risk-management lesson

Mark-to-market discipline controls default risk but can transform market risk into funding pressure.

Options

Call option

Gives the holder the right, but not the obligation, to buy the underlying at strike price K.

Put option

Gives the holder the right, but not the obligation, to sell the underlying at strike price K.

The option buyer pays a premium because the payoff is one-sided.

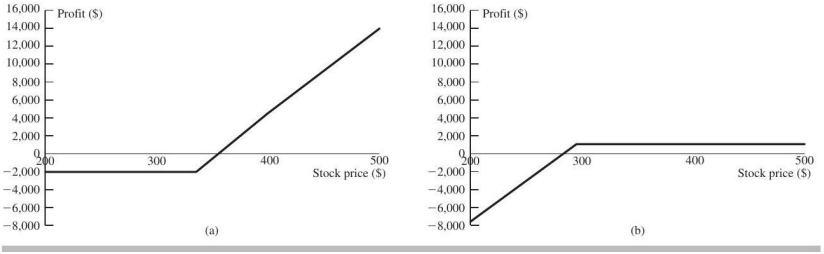
Option Payoffs

$$\text{Call payoff} = \max(S_T - K, 0) = (S_T - K)^+$$

$$\text{Put payoff} = \max(K - S_T, 0) = (K - S_T)^+$$

- Forward payoff is linear.
- Option payoff is kinked and convex.
- Convexity is why option pricing requires more machinery than forward pricing.

Option Profit Is Not Option Payoff



Premium matters

The buyer's payoff can be positive while profit is still negative if the payoff does not exceed the premium.

American Versus European Options

European

Exercise only at maturity.

American

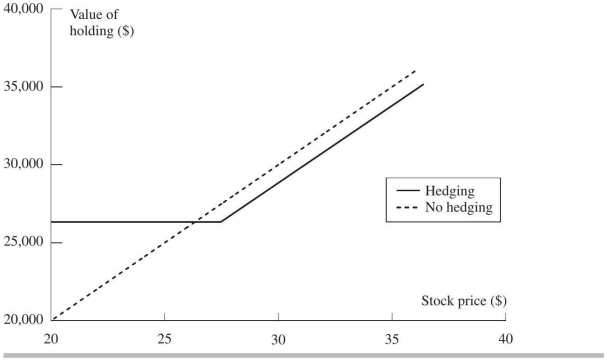
Exercise any time up to maturity.

Since American options give more flexibility,

$$C_A \geq C_E, \qquad P_A \geq P_E.$$

Hedgers

A hedger uses derivatives to reduce an existing exposure.



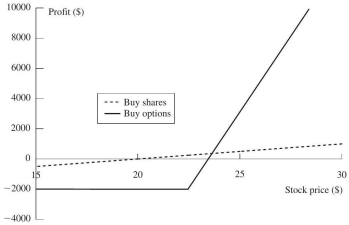
Evaluation

Do not judge the derivative leg alone. Judge the combined position.

Speculators

A speculator uses derivatives to take a view on market direction, volatility, spreads, or timing.

- Futures allow large notional exposure with margin rather than full purchase price.
- Options allow directional views with limited downside for buyers.
- Leverage makes both gains and losses arrive faster.



Arbitrageurs

An arbitrageur tries to lock in a profit by exploiting inconsistent prices.

True arbitrage

No net investment, no risk of loss, positive profit in at least one state.

Convergence trading is not true arbitrage

Many trades called arbitrage can lose money before prices converge. They are exposed to funding, liquidity, and model risk.

Why Study Mishaps in Lecture 1?

- Derivatives losses are rarely caused by equations alone.
- They usually combine leverage, weak controls, bad incentives, poor liquidity, and misunderstood models.
- The same instrument can be a hedge in one balance sheet and a speculative weapon in another.

Course discipline

Every pricing formula must be paired with its assumptions and its risk-management limits.

Common Failure Modes

1. Market moves against the position.
2. Position size is too large relative to capital.
3. Margin calls arrive before the strategy pays off.
4. Model prices or hedge ratios are wrong.
5. Traders can hide or misreport risk.
6. Risk limits are ignored after profitable breaches.
7. Incentives reward short-term gains and hide tail losses.

Risk Limits

A good risk limit answers four questions:

1. What exposure is being limited?
2. How is the exposure measured?
3. Who is notified when the limit is breached?
4. What action follows the breach?

Examples

Notional limits, VaR limits, stress-loss limits, Greek limits, stop-loss limits, maturity limits, counterparty limits, liquidity limits.

Front, Middle, and Back Office

Front office	Trades, manages client relationships, and generates revenue.
Middle office	Measures risk, validates prices, monitors limits, and challenges assumptions.
Back office	Confirms trades, settles cash flows, reconciles positions, and maintains records.

Control principle

The person taking risk should not be the only person measuring, confirming, and reporting that risk.

Model Risk

Models are useful simplifications, not reality.

- wrong model for the product;
- wrong parameters;
- ignored liquidity and transaction costs;
- unstable correlations in stress periods;
- untested extrapolation;
- users who do not understand assumptions.

Rule

A formula without assumptions is not knowledge.

What You Should Know After Lecture 1

- Derivatives are contracts with payoffs linked to underlying variables.
- Exchange and OTC markets solve different problems.
- Forwards create binding linear exposure.
- Futures add standardization, clearing, margin, and daily settlement.
- Options give rights rather than obligations and therefore require premiums.
- Hedging, speculation, arbitrage, and convergence trading are different activities.
- Risk management is institutional, not just mathematical.

Practice Before Next Class

1. Draw long and short forward payoff diagrams.
2. Compute the no-arbitrage forward price for a non-dividend-paying stock.
3. Compute daily futures P&L for long and short positions.
4. Explain why margin is collateral rather than purchase price.
5. Write call and put payoffs using positive-part notation.
6. Identify whether a trade is hedging, speculation, arbitrage, or convergence trading.
7. Pick one derivatives mishap and identify the risk-management failures.

Next class: hedging with futures and forward/futures prices.